

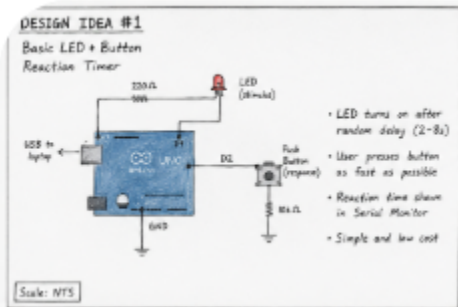
Element D V1

MyDesign Portfolio

Element D Initial Template AT - 15 points

Element D: Design Concept Generation, Analysis, and Selection

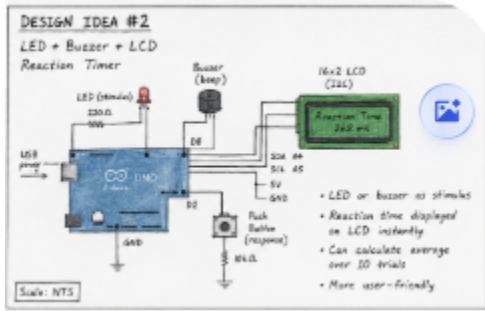
Design Idea Generation (D1)

Design Idea #1	Arduino
Date:	1/20/2026
Description:	<i>Basic Arduino reaction timer using LED stimulus + push button response + serial monitor output. The system flashes a single LED after a random delay and measures reaction time using millis().</i>
Drawing that includes distinguishing information such as scale, dimensions, and materials:	

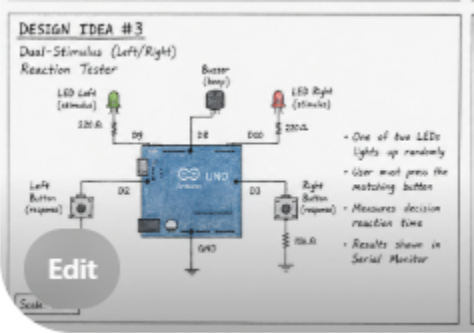


Element D V1

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Design Idea #2	Improved Arduino
Date:	2/10/2026
Description:	Improved reaction tester with LED + buzzer + LCD display (16x2). Reaction time shown instantly on screen and stored for averaging. + Buzzer
Drawing that includes distinguishing information such as scale, dimensions, and materials:	 DESIGN IDEA #2 LED + Buzzer + LCD Reaction Timer Scale: NTS • LED or Buzzer as stimulus • Reaction time displayed on LCD instantly • Can calculate average over 10 trials • More user-friendly

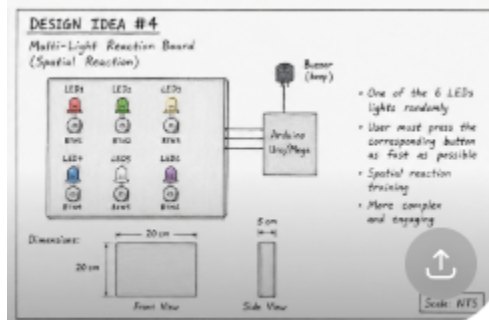
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Design Idea #3	Dual Arduino
Date:	3/5/2026
Description:	<i>Dual-stimulus system testing decision reaction time: two LEDs (left/right) light randomly and user must press matching button.</i>
Drawing that includes distinguishing information such as scale, dimensions, and materials:	

Design Idea #4	Sports Style
Date:	4/3/2026
Description:	<i>Sports-style reaction board inspired by FitLight: multiple LEDs arranged in grid, testing spatial reaction speed.</i>

Element D V1

Drawing that includes distinguishing information such as scale, dimensions, and materials:



Design Idea Comparison and Selection (D1, D2)

(List your requirements for Element C below.)	Weight (multiplied by each design score)	Design 1	Design 2	Design 3	Design 4	Design 5
Accuracy	5	4	5	4	4	
Random delay control	4	5	5	5	5	
Trial averaging	4	3	5	3	4	

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Ease of use	3	5	5	4	3	
Cost Efficiency	5	5	5	3	4	
Total score	3	5	5	5	4	

Weighted Design Matrix

Requirement	Weight	Idea 1	Idea 2	Idea 3	Idea 4	Idea 5
Accuracy (± 5 ms)	5	4	5	4	4	5
Random delay control	4	5	5	5	5	5
10-trial averaging	4	3	5	3	4	5
Ease of use	3	5	5	4	3	3
Cost efficiency	5	5	3	4	2	2
Immediate feedback	4	3	5	3	4	5
Physical realism (button input)	3	5	5	5	4	5

Design Idea #2 was selected because it best satisfies all high-priority design requirements from Element C. It achieves high measurement accuracy (millisecond-level via Arduino timing functions) while also providing immediate user feedback through an LCD display, which improves usability compared to serial-only systems.

It also successfully integrates: randomized stimulus delay (2-8 seconds) physical button response (removes touchscreen error) multi-trial averaging (improves reliability) dual stimulus options (LED + buzzer for accessibility) Compared to simpler systems (Idea 1), it significantly improves usability and feedback. Compared to more

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complex systems (Ideas 4 and 5), it maintains lower cost and simpler implementation, making it more suitable for a working prototype while still meeting scientific validity requirements.

Design Blueprint (D4)

20/05/2026

Item	Cost each	# needed	Total
Arduino	12	1	12
Push button	0.5	1	0.5
Led + Buzzer	1.2	1	1.2
Total Cost			13.7

Plan of Action (D3)

System Overview

Arduino-based reaction time tester using: LED (visual stimulus) Buzzer (audio stimulus option) Push button (response input) LAPTOP DISPLAY

Arduino Uno Connections LED → Digital Pin 9 (via 220Ω resistor) Buzzer → Digital Pin 8 Push Button → Digital Pin 2 (with 10kΩ pull-down resistor) LCD (I2C): SDA → A4 SCL → A5 VCC → 5V GND → GND LED → Digital Pin 9 (via 220Ω resistor) Buzzer → Digital Pin 8 Push Button → Digital Pin 2 (with 10kΩ pull-down resistor) LCD (I2C): SDA → A4 SCL → A5 VCC → 5V GND → GND

Manufacturing Methods Step 1: Build circuit on breadboard and test Arduino code Step 2: Transfer to permanent wiring (solder or perfboard) Step 3: Cut enclosure openings (LCD, LED, button) Step 4: Secure components using screws or hot glue Step 5: Final assembly and system calibration

Date of Iteration: 4/28/2026

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Description of Iteration: Having some error with the matlab integration STATUS INCOMPLETE

Date of Iteration: 5/04/2026

Description of Iteration: Successfully got the program to test reaction times, but no buzzer and no average STATUS INCOMPLETE

Date of Iteration: 5/05/2026

Description of Iteration: Successfully met all the requirements STATUS COMPLETE

Rubric

Element D: Design Concept Generation, Analysis and Selection						
	5	4	3	2	1	0

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1. Process for Generating & Comparing Possible Solutions. The Process for Generating Possible Solutions _____.	was <i>comprehensive, iterative, and consistently</i> defensible (viable design highly likely)	was <i>thorough, iterative, and</i> generally defensible (viable design likely)	was <i>adequate, generally</i> iterative and defensible (<i>viable design is possible</i>)	was <i>partial or overly general and only somewhat</i> iterative and/or defensible (raising issues with the design viability)	was <i>incomplete or minimally iterative and/or defensible</i> (raising issues with the design viability)	<i>was not evident that there was an attempt</i>
2. Design Solution Choice. The Design Solution chosen _____.	was <i>well-justified</i> and demonstrated attention to <i>all</i> design requirements	was <i>justified</i> and demonstrated attention to <i>most if not all</i> , design requirements	was <i>explained with reference to at least some</i> design requirements	was <i>not sufficiently explained with reference to</i> design requirements	<i>lacked support related to</i> design requirements	<i>was not attempted or evidence not provided</i>
3. Plan of Action. The plan of action _____.	<i>has considerable merit and would easily</i> support repetition and testing for effectiveness by others	<i>would</i> support repetition and testing for effectiveness by others	<i>might not clearly or fully support repetition and testing for effectiveness by others</i>	<i>has insufficient detail to allow for testing for replication of results</i>	<i>is lacking to the point that the design solution cannot be tested</i>	<i>cannot be executed or does not exist</i>